



SPCapstone002

Presentation Report

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Abstract

This study delves into real estate analysis within the Philippines by thoroughly cleaning a scraped dataset obtained from Lamudi via Kaggle. Despite challenges such as missing values and extreme outliers, the employed predictive models, particularly Random Forest Regression and XGBoost, offer stakeholders in the local housing market an additional tool for making well-informed decisions. While acknowledging the constraints of the limited dataset, this work signifies an initial step in comprehending and forecasting house prices in the Philippines. The recommendation for future initiatives involves enhancing data collection through the incorporation of timestamps. This approach facilitates ongoing monitoring of housing price trends, allowing for the application of advanced analytical techniques. By doing so, the effectiveness of machine learning models is expected to improve. Additionally, it is recommended to develop a detailed and accurate geospatial visualization in conjunction with property prices, contributing to a comprehensive understanding of the real estate landscape.

A. Introduction

Importance of Predicting House Prices

The ability to predict house prices holds immense significance in real estate markets, offering numerous benefits to various stakeholders. Whether it's homebuyers, sellers, developers, or policymakers, accurate predictions contribute to informed decision-making.

Dataset

The analysis focuses on the "House Prices in the Philippines" dataset retrieved from Kaggle, a comprehensive collection of housing market data across various regions in the country. Scraped from lamudi. This dataset encompasses information on key variables such as property category, size, location, price among others.

Significance

The unpredictability of house prices in the Philippines poses challenges for both individuals and the real estate industry. By addressing this issue through predictive analytics, the proposed solution aims to **facilitate informed investments** where Investors, both individual and institutional, can make more informed decisions, fostering a more robust and sustainable real estate market.

B. Statement of the Problem

The housing market in the Philippines is characterized by a lack of predictability, creating challenges for homebuyers, sellers, developers, and policymakers. The inconsistent nature of house price fluctuations makes it difficult for individuals and entities to make informed decisions regarding property transactions.

C. Objectives

The primary objective of this analysis is to address the issue of unpredictability in house prices in the Philippines by developing a robust predictive model. Through the utilization of relevant features within the "House Prices in the Philippines" dataset.

To implement and develop a predictive model through advanced data analysis and machine learning for property prices.

To identify and determine the key features within the dataset that significantly influence house prices. This involves a thorough analysis of variables.

To provide homebuyers, sellers, and developers with a tool that enhances their ability to make informed decisions based on found key features for an appropriate property price.

D. Analysis

Data Collection

Source:

- The dataset, titled "House Prices in the Philippines," retrieved from [Kaggle](#) and its values was scraped from online real estate listings, specifically www.lamudi.com.ph.

Size:

- The dataset comprises a total of 2983 entries, with each row representing a unique property listing.

Variables:

1. **Description:** Brief description of the property.
2. **Location:** Geographical location of the property.
3. **Category:** Type of property (e.g., Condominium, House and Lot).
4. **Sub-Category:** Further classification of the property type.
5. **Land Size(M2):** Size of the land in square meters.
6. **Building Size(M2):** Size of the building in square meters.
7. **Car Space:** Number of available parking spaces.
8. **Bedrooms:** Number of bedrooms in the property.
9. **Bathrooms:** Number of bathrooms in the property.
10. **Longitude:** Geographic coordinate of the property's location (longitude).

11. **Latitude:** Geographic coordinate of the property's location (latitude).

12. **Condominium Name:** Name of the condominium if applicable.

13. **Price(Php):** Property price in Philippine Pesos.

14. **Link:** URL link to the property listing.

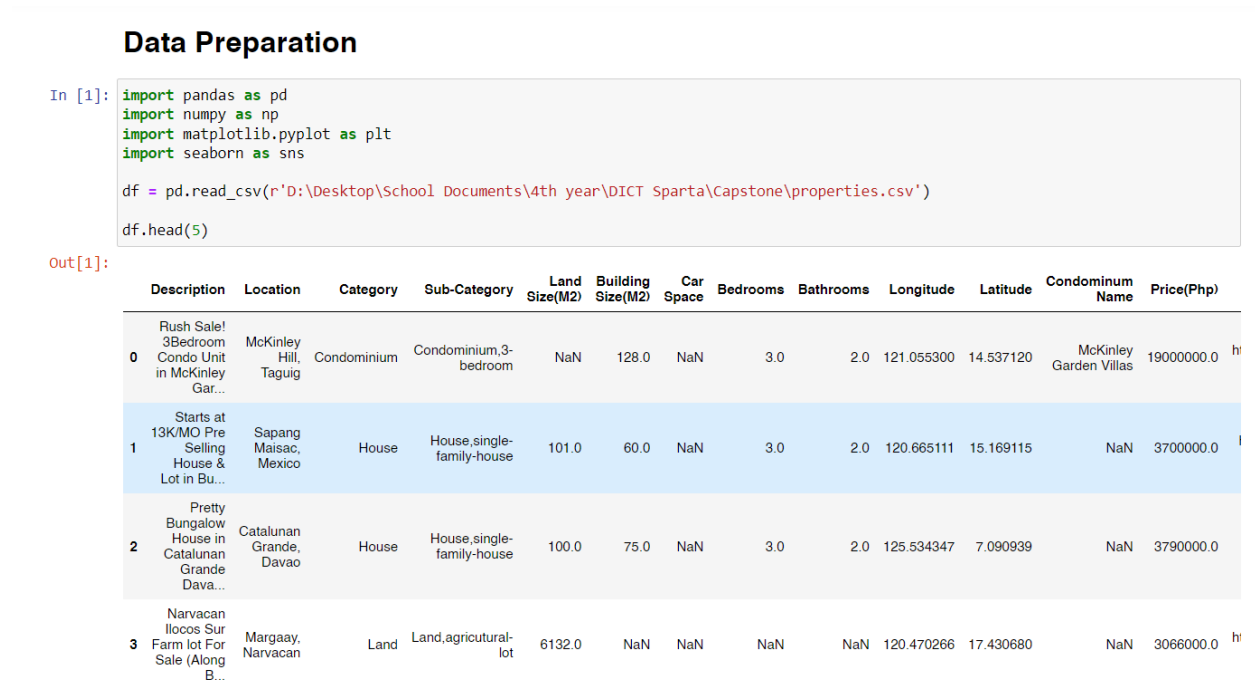


Figure 1 – Imported Data.

Figure 1 illustrates the dataset that has been imported, presenting a visual representation of its contents and associated values.

Exploration of the Dataset Structure

Upon importing the dataset, a thorough exploration was conducted to understand its structure and characteristics. The dataset, sourced from real estate listings, contains various features representing key attributes of properties in the Philippines.

Cleaning and Preprocessing

To prepare the dataset for predictive modeling, the following steps were undertaken:

1. Limiting Price Values:

The dataset's scope was refined by restricting the analysis to properties with a price value less than 100 million PHP. This was done to mitigate the impact of outliers on the predictive model.

2. Column Selection:

Columns deemed irrelevant for forecasting, such as 'Description,' 'Location,' 'Link,' 'Sub-Category,' and 'Condominium Name,' were dropped. This streamlines the dataset, focusing only on the essential features for the predictive model.

```
In [6]: # I'll be limiting the scope of the dataset to only Price valued at less than 100M to limit outliers
df = df[df['Price(PHP)'] <= 100000000]

In [7]: # Dropping these columns as it is not relevant for our forecasting.
df.drop(['Description', 'Location', 'Link', 'Sub-Category', 'Condominium Name'], axis=1, inplace=True)

In [8]: # Since having no car space in the dataset is equal to none, it is necessary to change it to numerical 0
df['Car Space'].fillna(0, inplace=True)
```

Figure 2 – Code snippet for Step 1 and 2.

3. Handling 'Car Space' Values:

The 'Car Space' column exhibited instances where the value 0 was represented as 'None.' To address this, the 'None' entries were converted to numerical 0, ensuring consistency in the dataset.

```
In [8]: # Since having no car space in the dataset is equal to none, it is necessary to change it to numerical 0
df['Car Space'].fillna(0, inplace=True)
```

Figure 3 – Code snippet for Step 3.

4. Missing Value Imputation:

Given the constrained nature of the dataset, preserving its integrity involves identifying and rectifying missing values. Recognizing the significance of land size as a pivotal feature, the decision was made to eliminate missing entries within that column. The MICE (Multiple Imputation by Chained Equations) method from the miceforest library was employed for this purpose, leveraging its capability to impute missing values by considering the interrelationships among variables, thereby augmenting the dataset's completeness.

```
Check for missing values

In [9]: print("Missing Values by Column")
print("-"*30)
print(df.isna().sum())
print("-"*30)
print("TOTAL MISSING VALUES:", df.isna().sum().sum())

Missing Values by Column
-----
Category                0
Land Size(M2)           828
Building Size(M2)       584
Car Space                0
Bedrooms                675
Bathrooms               720
Longitude                1
Latitude                 1
Price(PHP)              0
dtype: int64
-----
TOTAL MISSING VALUES: 2809

In [10]: df.dropna(subset=['Land Size(M2)', 'Longitude', 'Latitude'], inplace=True)

In [11]: #Fill missing values using MICE (Multiple Imputation by Chained Equations) method from the miceforest library
from miceforest import ImputationKernel

kernel = ImputationKernel(df[['Building Size(M2)', 'Bedrooms', 'Bathrooms']])
kernel.mice(5, verbose=True)
df[['Building Size(M2)', 'Bedrooms', 'Bathrooms']] = kernel.complete_data()
kernel.complete_data()

Initialized logger with name mice 1-5
Dataset 0
1 | Building Size(M2) | Bedrooms | Bathrooms
2 | Building Size(M2) | Bedrooms | Bathrooms
3 | Building Size(M2) | Bedrooms | Bathrooms
4 | Building Size(M2) | Bedrooms | Bathrooms
5 | Building Size(M2) | Bedrooms | Bathrooms

Out[11]:
   Building Size(M2)  Bedrooms  Bathrooms
1          60.000      3.000      2.000
2          75.000      3.000      2.000
3         230.000      3.000      3.000
4          100.630      3.000      2.000
6          850.000      5.000      5.000
...
2994         500.000      4.000      4.000
2995         700.000      5.000      4.000
2996         145.000      4.000      3.000
2997         350.000      4.000      4.000
2999         236.000      6.000      6.000
```

Figure 4 – Snippet code for Step 4.

5. Outlier Removal:

Outliers in numerical columns, including 'Land Size(M2)', 'Building Size(M2)', 'Car Space', 'Bedrooms', 'Bathrooms', and 'Price(Php)', were identified and removed using z-score normalization. This step ensures that extreme values do not unduly influence the predictive model.

```
Remove Outliers from the Dataset

In [15]: numerical_features = ['Land Size(M2)', 'Building Size(M2)', 'Car Space', 'Bedrooms', 'Bathrooms', 'Price(Php)']

In [16]: from scipy.stats import zscore
         z_threshold = 1
         # Calculate z-scores for the specified columns
         z_scores_price = zscore(df['Price(Php)'])
         # Identify and remove outliers
         df = df[(abs(z_scores_price) <= z_threshold)]

In [17]: df.describe()
Out[17]:
```

	Land Size(M2)	Building Size(M2)	Car Space	Bedrooms	Bathrooms	Longitude	Latitude	Price(Php)
count	1511.000	1511.000	1511.000	1511.000	1511.000	1511.000	1511.000	1511.000
mean	813.625	235.473	0.863	3.634	3.174	121.162	14.041	14065078.128
std	4320.281	257.215	1.791	1.283	1.612	2.903	1.576	12384531.175
min	0.000	0.000	0.000	0.000	0.000	15.113	6.160	15000.000
25%	100.000	88.000	0.000	3.000	2.000	120.961	14.192	4833625.000
50%	189.000	166.000	0.000	4.000	3.000	121.025	14.423	9450000.000
75%	347.000	300.000	2.000	4.000	4.000	121.108	14.603	20750000.000
max	98549.000	3015.000	50.000	11.000	17.000	126.780	18.275	48263000.000

Data is now cleaned with no missing values, with a final instances of 1511

Figure 4 – Snippet code for Step 5.

These preprocessing steps are crucial in refining the dataset, enhancing its quality, and preparing it for the succeeding stages of analysis and modeling. The cleaned and processed dataset serves as the foundation for the development of a predictive model for house prices. The following sections will delve into the specifics of feature selection, model development, and result interpretation.

Feature Selection

To develop an effective predictive model for house prices, a thorough process of feature selection was undertaken. The selection of relevant features was guided by a combination of domain knowledge and statistical analysis.

The following methods were employed to identify and justify the inclusion of specific features:

a. Data Correlation Matrix and Heatmap

A correlation matrix and heatmap were generated to visualize the relationships between different features in the dataset. This provided insights into the strength and direction of correlations, guiding the selection of features with significant associations with house prices.

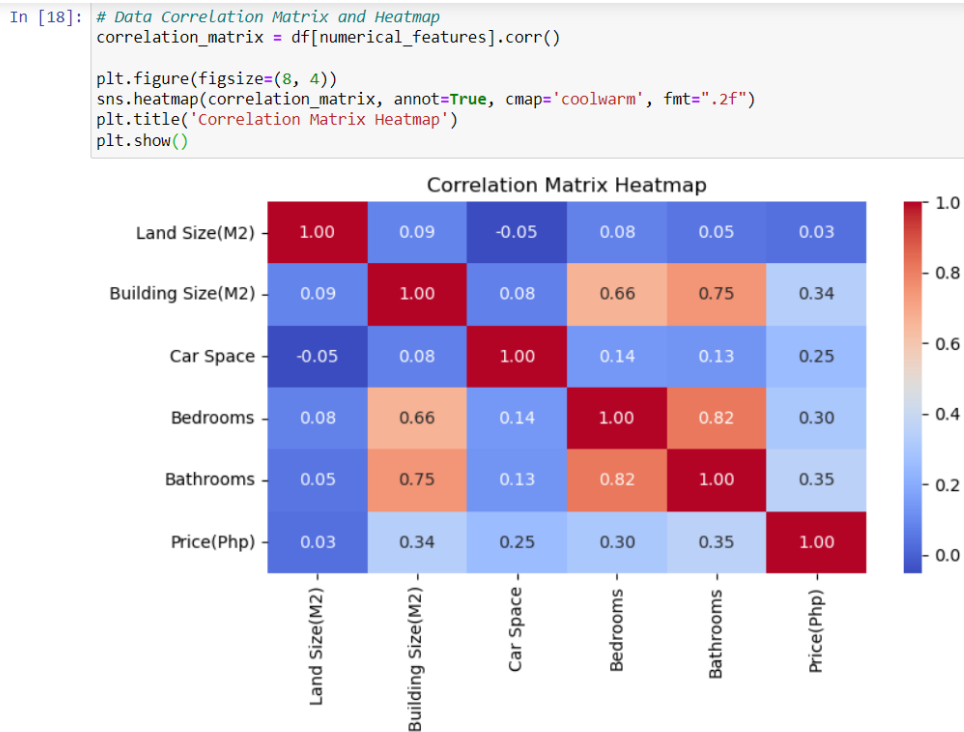


Figure 5 – Correlation Matrix Heatmap.

b. Pair plot of Numerical Columns

A pair plot was created for numerical columns, including 'Building Size(M2)', 'Car Space,' 'Bedrooms,' 'Bathrooms,' and 'Price(Php).' This graphical representation facilitated the examination of pairwise relationships, aiding in the identification of features that exhibit patterns relevant to price prediction.

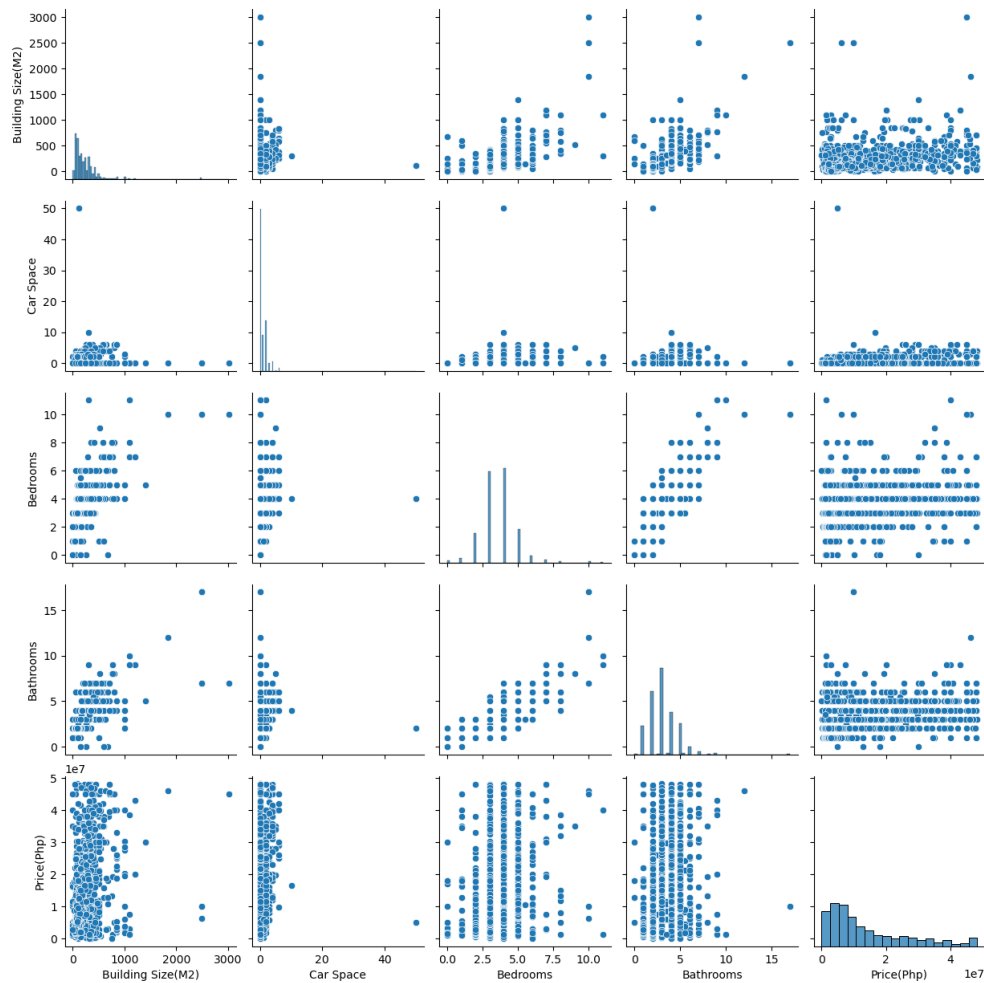


Figure 6 – Pair plot of Numerical Columns.

c. Histogram of Price

A price histogram was generated to analyze the distribution patterns and frequencies of price values within the dataset. This visual representation offers insights into how prices are distributed across the properties, allowing for a comprehensive understanding of the pricing landscape.

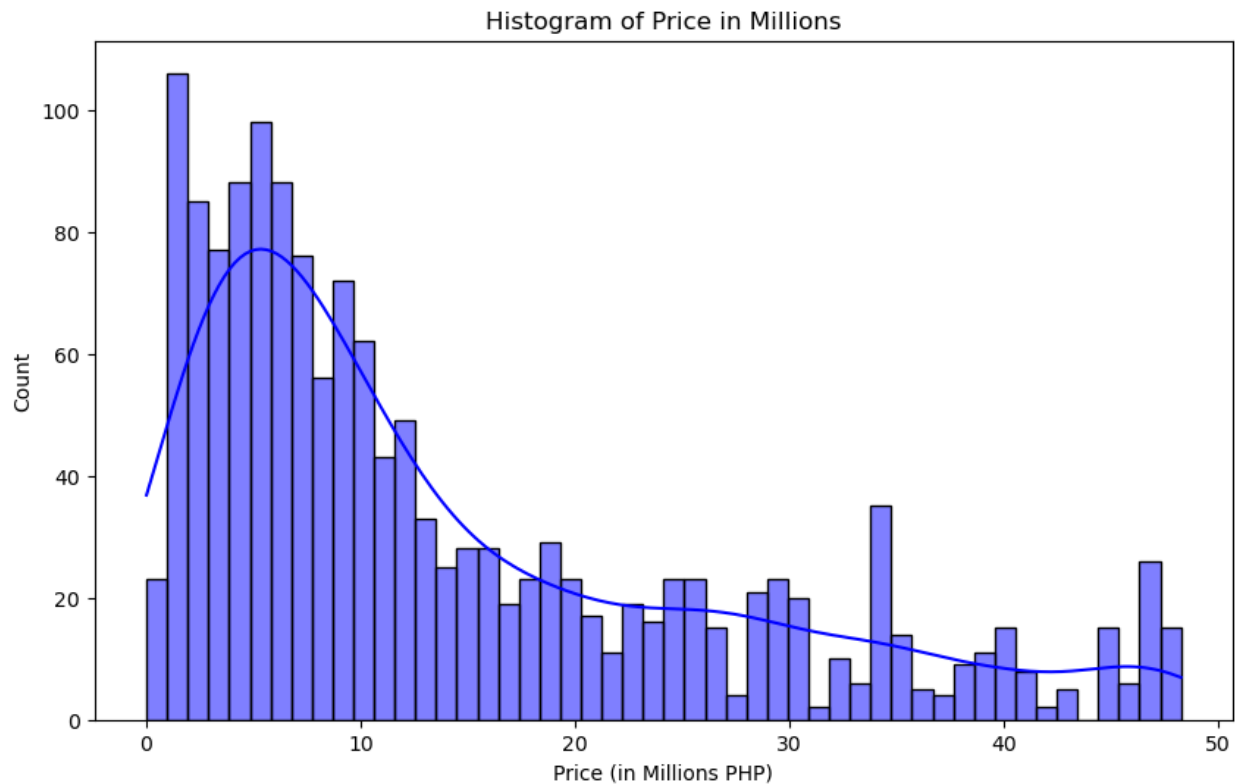


Figure 7 – Histogram of Column Price(Php).

d. Boxplot of Numerical Columns:

Boxplots were constructed for numerical columns to visualize the spread and central tendency of each variable. This method facilitated the identification of potential outliers and the impact of different numerical features on house prices.

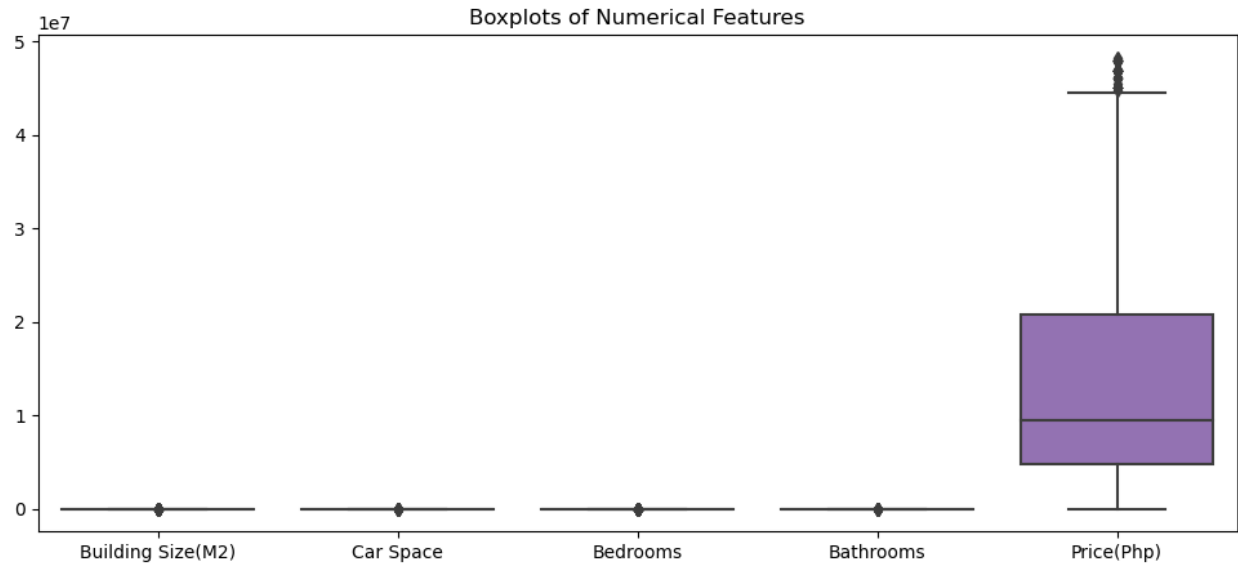


Figure 7 – Boxplots of Numerical Features.

e. Geospatial Visualization

Geospatial visualization using geopandas, along with an OpenStreetMap basemap of the Philippines through contextily, was performed. The Price(Php) color scale on the map provided a spatial understanding of how property prices vary across different geographic locations. It can be noted that most of the listings in the dataset are in Central Luzon.

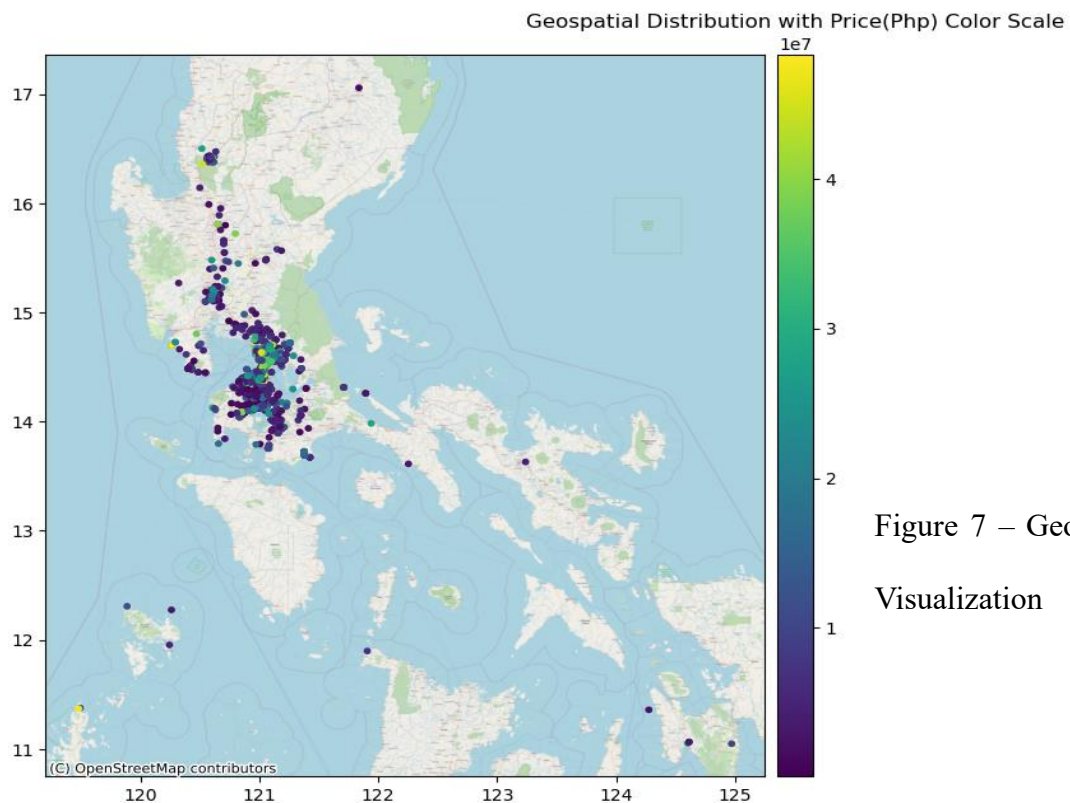


Figure 7 – Geospatial Visualization

Model Selection

Model Choices and Rationale

The process of selecting an appropriate predictive model for the dataset involved the consideration of various algorithms. After preprocessing steps, including X, y split, One-Hot encoding for the 'Category' column, and standardization using a standard scaler, the following models were chosen:

a. Linear Regression

Linear regression was chosen as it provides a simple yet interpretable model. Given the nature of the housing market, where linear relationships may exist between certain features and house prices, linear regression serves as a baseline model for comparison.

b. Random Forest Regression

Random Forest Regression was selected due to its ability to handle complex relationships and non-linear patterns in the data. The ensemble of decision trees helps capture intricate feature interactions, making it suitable for real estate datasets with multifaceted dependencies.

c. XGBoost Regressor

XGBoost Regressor is an ensemble learning algorithm known for its efficiency and effectiveness in handling diverse datasets. It was chosen for its robustness in managing both linear and non-linear relationships, making it suitable for the diverse features present in the dataset.

d. Support Vector Machines (SVM)

SVM was considered for its capability to handle high-dimensional datasets and capture complex relationships. SVM can uncover intricate patterns in the data, and its versatility makes it well-suited for real-world applications with various feature types.

e. Neural Network

A simple Neural network were included for their ability to learn intricate representations and patterns in data. The complexity of the housing market data justifies the exploration of neural networks to capture non-linear relationships.

f. Elastic Net

Elastic Net combines the strengths of both L1 and L2 regularization, making it suitable for real estate datasets. It helps prevent overfitting, contributing to the robustness of the model.

The chosen models offer a diverse set of tools to capture the nuances of house price prediction, considering both linear and non-linear relationships within the dataset. The subsequent sections of the report will delve into the training, evaluation, and interpretation of results for each selected model, providing a comprehensive understanding of their performance in predicting house prices in the Philippines.

Dataset Splitting

To ensure evaluation, the dataset was split into training and testing sets using the sklearn `train_test_split` method, with a ratio of 70% for training and 30% for testing.

Train-Test Split

```
In [54]: from sklearn.model_selection import train_test_split, cross_val_score
         x_train, x_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)
```

Figure 8 – Code Snippet for Dataset Splitting

Model Training and Evaluation Metrics

The selected models Linear Regression, Random Forest Regression, XGBoost Regressor, Support Vector Machines (SVM), Neural Network, and Elastic Net were trained and evaluated using the following metrics:

a. Mean Absolute Error (MAE)

MAE measures the average absolute differences between predicted and actual values, providing an indication of the model's accuracy.

b. Mean Squared Error (MSE)

MSE calculates the average squared differences between predicted and actual values, penalizing larger errors more heavily.

c. Root Mean Squared Error (RMSE)

RMSE is the square root of the MSE, providing an interpretable measure of the average magnitude of errors.

d. R-squared (R^2)

R-squared quantifies the proportion of the variance in the dependent variable that is predictable from the independent variables. It ranges from 0 to 1, where 1 indicates a perfect fit.

e. Root Mean Squared Error Cross-Validation (RMSE CV)

Utilizes cross-validation to assess the model's predictive performance. It provides an average RMSE across multiple folds, enhancing the robustness of the evaluation.

The models were trained on the training set, and their performance was assessed using the testing set. The evaluation metrics offer a comprehensive view of how well each model generalizes to unseen data.

E. Results and Discussion

The results from each model's evaluation metrics will be systematically analyzed and compared. This comparative analysis will provide insights into the strengths and weaknesses of each model, guiding the selection of the most effective approach for predicting house prices in the Philippines.

Model Performance Metrics

The performance of various predictive models for house prices in the Philippines was assessed using multiple evaluation metrics. The results are summarized in the figure below:

```
In [64]: models.sort_values(by="RMSE (Cross-Validation)")
```

```
Out[64]:
```

	Model	MAE	MSE	RMSE	R2 Score	RMSE (Cross-Validation)
1	Random Forest Regression	4643167.565	64660797105257.055	8041193.761	0.594	7959845.593
2	XGBoost	4933215.286	77782704754898.250	8819450.366	0.512	8584364.247
4	Elastic Net	8800796.324	135509370674573.125	11640849.225	0.150	11446819.444
0	LinearRegression	8459557.067	132056650195312.953	11491590.412	0.172	12008845.570
3	Support Vector Machines (SVM)	9199549.851	180996934966939.125	13453510.136	-0.135	13188460.130

Figure 9 – Code snippet and table of summarized result of models

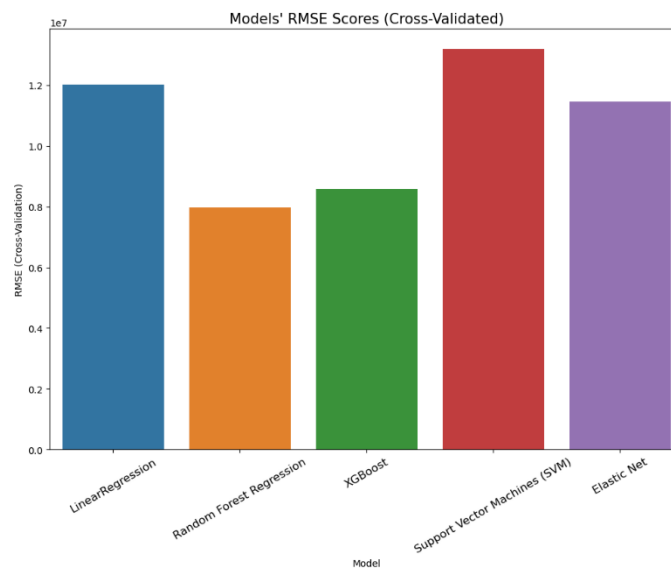


Figure 10 – Bar plot of Model's RMSE Scores (Cross-Validated)

Feature Importance Analysis

Given that Random Forest Regression and XGBoost demonstrated relatively higher R² scores, feature importance analysis was performed exclusively on these two models:

a. Random Forest Regression:

Top 3 Feature Importance:

1. Land Size
2. Building Size
3. Category 'Land'

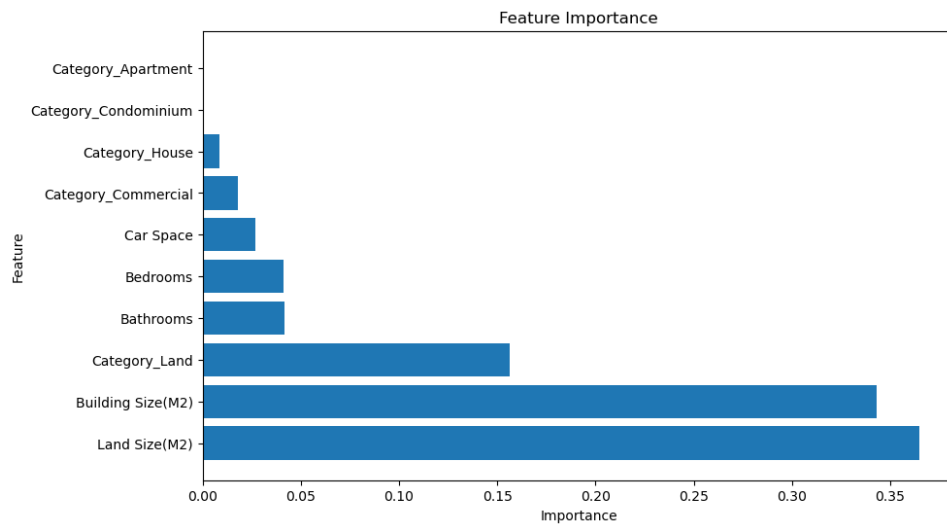


Figure 11 – Feature importance visualization for Random Forest Regression

b. XGBoost:

Top 3 Feature Importance:

1. Category 'Land'
2. Category 'Commercial'

3. Building Size

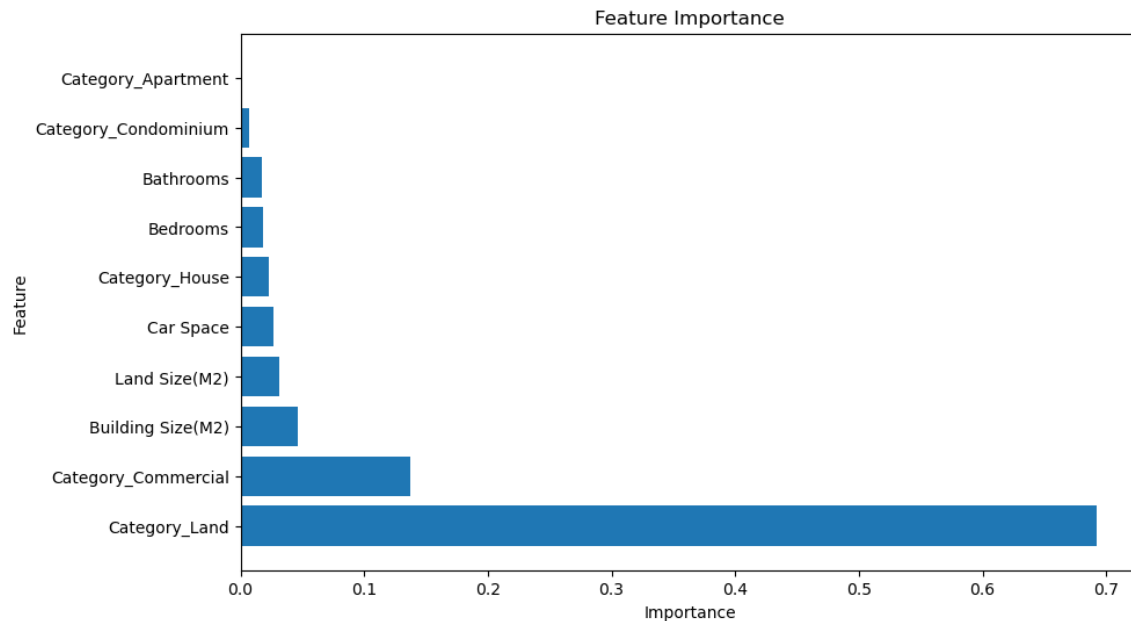


Figure 12- Feature importance visualization for XGBoost Regressor

Random Forest and XGBoost Significance

Random Forest Regression and XGBoost outperformed other models, achieving R2 scores above 0.50. These models demonstrated a better ability to capture the complex relationships within the dataset.

Feature Importance Insights

In Random Forest Regression, land size, building size, and having a category of 'land' type were identified as the top three influential features. For XGBoost, the most crucial features were 'Category_Land,' 'Category_Commercial,' and 'Building Size.'

Significance of Features

The results emphasize the importance of spatial features (land size), physical attributes (building size), and property category in predicting house prices. These findings align with real-world expectations, where location, size, and type significantly influence property values.

F. Conclusion

After thoroughly cleaning the scraped dataset obtained from Lamudi via Kaggle, a significant number of entries were excluded due to either missing values or extreme outliers, further constraining the already limited dataset pertaining to housing prices in the Philippines. Nevertheless, despite these challenges, this marks an initial step in real estate analysis within the Philippines. The employed predictive models, particularly Random Forest Regression and XGBoost, provide stakeholders in the local housing market with an additional tool for making well-informed decisions. The analysis highlights the pivotal role of advanced predictive modeling in comprehending and forecasting house prices in the Philippines.

G. Recommendation

The leading models, Random Forest Regression and XGBoost, achieved R^2 scores slightly above 0.50, a performance likely influenced by the limitations of the dataset. For future initiatives, it is recommended to enhance data collection by incorporating timestamps, facilitating continual monitoring of housing price trends in the Philippines. This strategy enables the application of advanced analytical techniques, improving the efficacy of machine learning models and resulting in enhanced outcomes. Furthermore, an effective solution includes the use of a more detailed and accurate geospatial visualization in conjunction with property prices.

References

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